



**Tech
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**From sustainability policy to
strategic innovation: analyzing
circular maturity and business
transformation in Colombia's
plastic value chain**

**ECHO 2026 – Sustainability and the
Circular Future with UniGR-CIRKLA**

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UniGR | Centre for Circular Economy
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Plastic Problem: View from Colombia

In Colombia, while a robust regulatory landscape has evolved, actual value chains remain highly linear.

Structural Barriers:

1. Underfunded technology transfer offices
2. Misalignment between academic outputs and industry needs
3. Industry needs hinder effective collaboration.



Goal & Approach

Identification of the barriers and lock-ins to Technology Transfer within Colombia's plastics value chain.

A significant gap persists between knowledge generation and its effective implementation in the productive sector

The study employs a mixed qualitative-quantitative approach that includes an assessment of the productive sector through surveys of 92 companies and semi-structured interviews with key stakeholders



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Acoplásticos

Methodology

Phase I Documentary Analysis

**Legal framework
of CE within
Colombian
context**



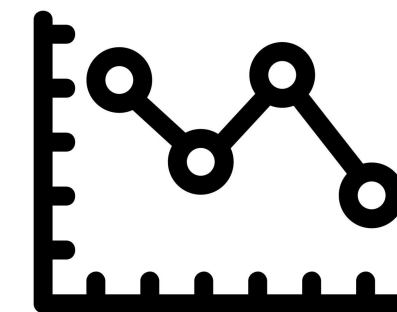
Phase II Product sector diagnosis (Interviews and surveys)

**Exploratory
survey to
companies and
semi-structured
interviews**



Phase III Evaluation of Circular Maturity (survey)

**Innovation index
determined
based on
companies
self-diagnosis**



Phase IV Validation and strategy design proposals

**48-h co-creation
workshop with
students and
industry experts**



Key finding: The sustainability paradox

There is a significant gap between the formalization of corporate sustainability policies and the existence of structured innovation processes.

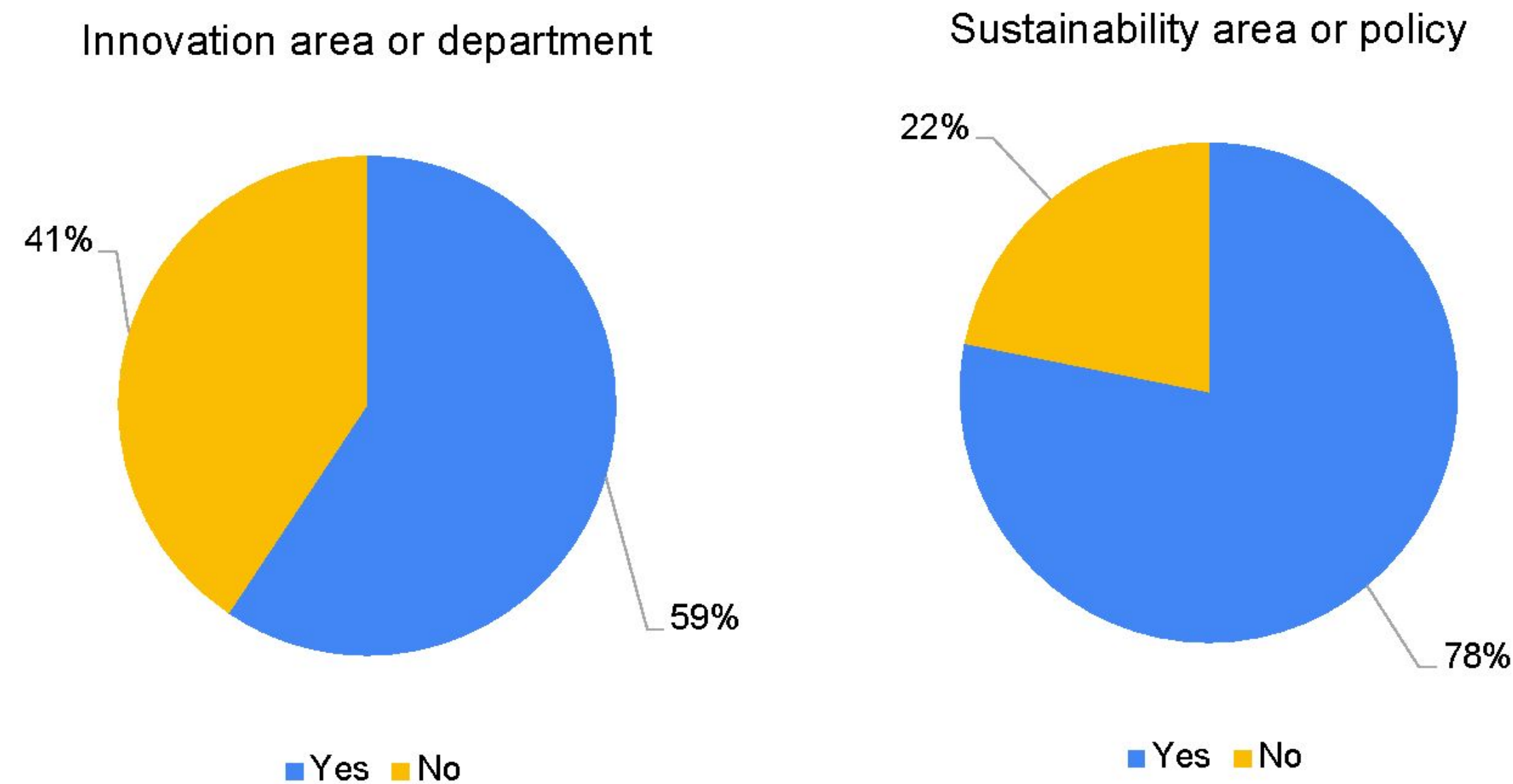
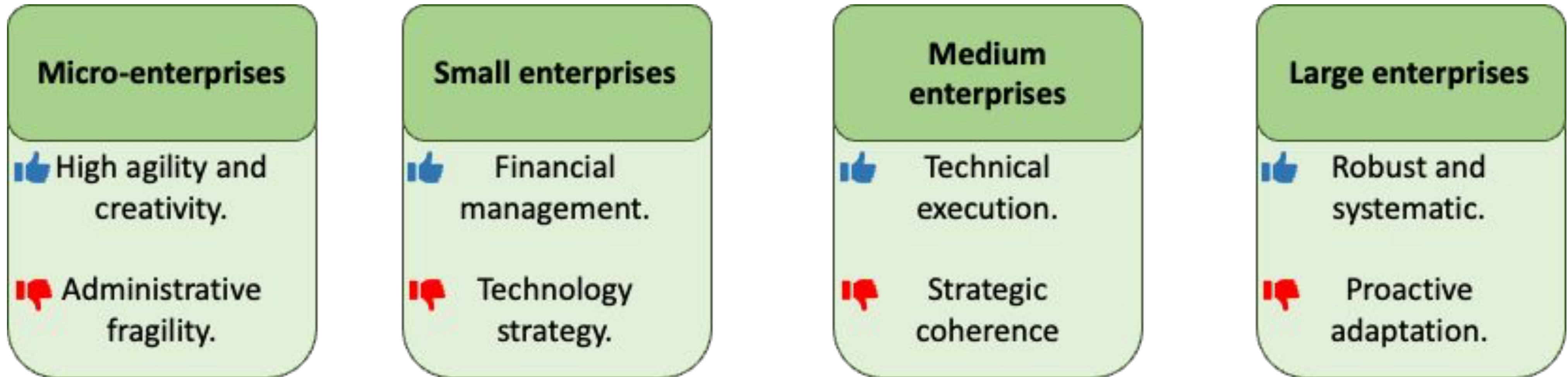


Figure 1. Companies that reported having an innovation area or department, and a sustainability area or policy.

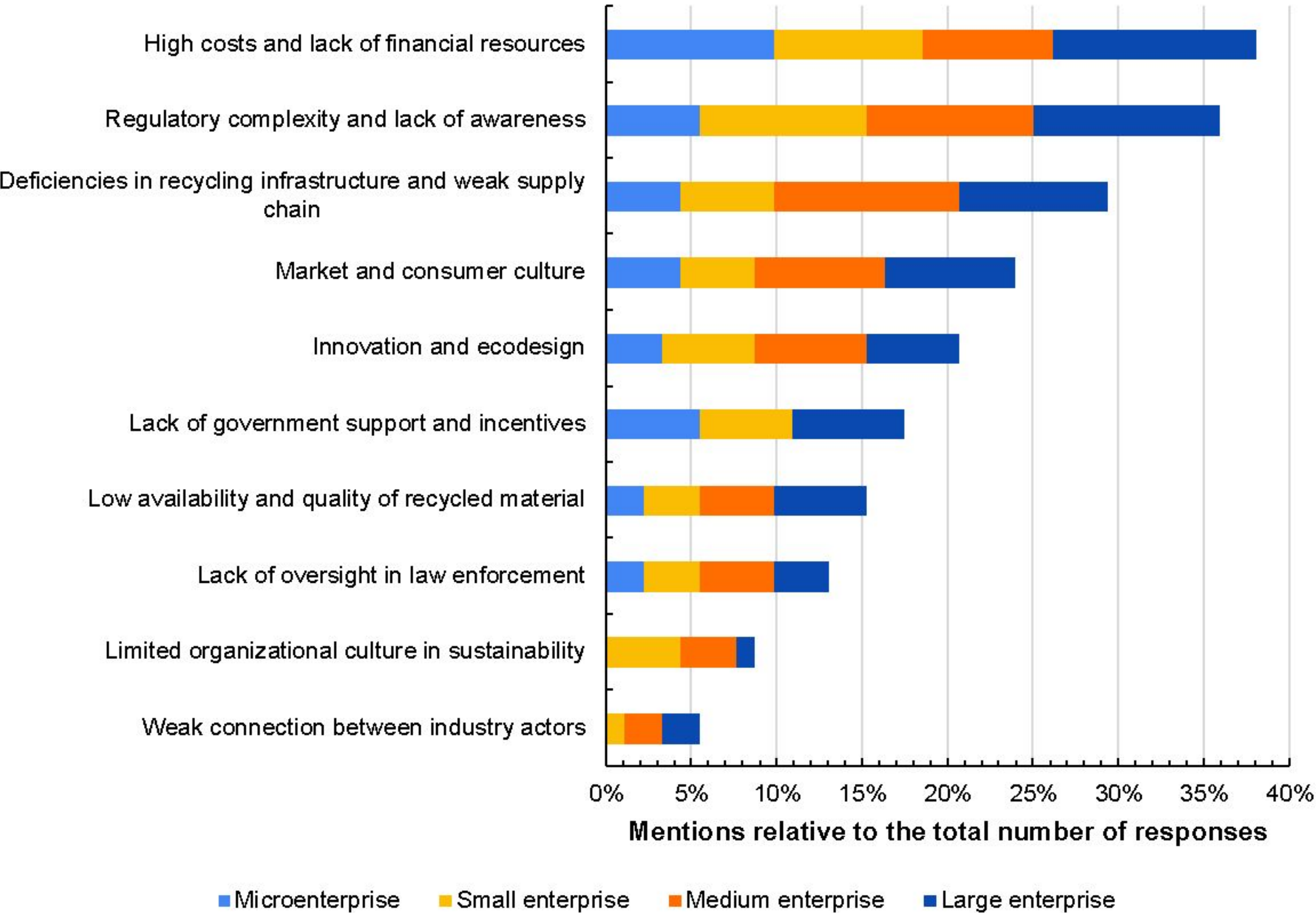
Key finding: Asymmetries in circular maturity

Different maturity profiles towards circular economy practices by enterprise size.



Key finding: Systemic Barriers

Figure 2. Most frequently reported barriers to compliance with plastics and circular economy regulations, and their relevance according to company size.



Key finding: Academia - Industry Alignment

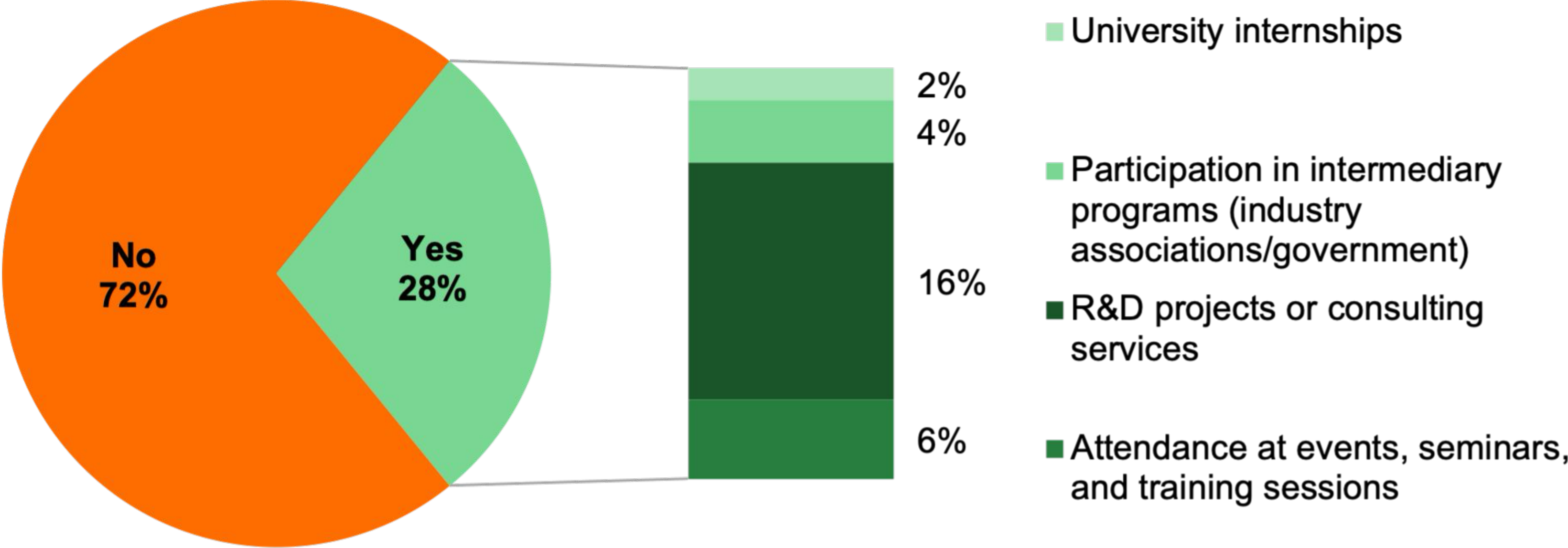
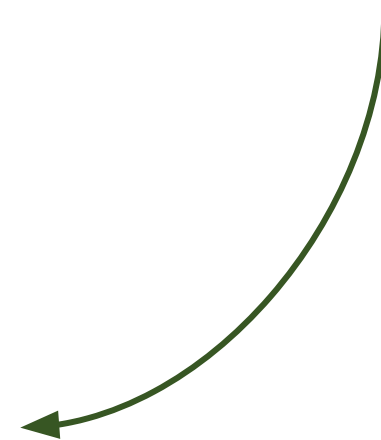


Figure 3. Companies reporting relationships with academia and types of collaboration.

Opportunities : Academia - Industry

Emerging opportunities exist in product ecodesign and digital value chain integration to transform plastic from an environmental liability into a high-value resource.

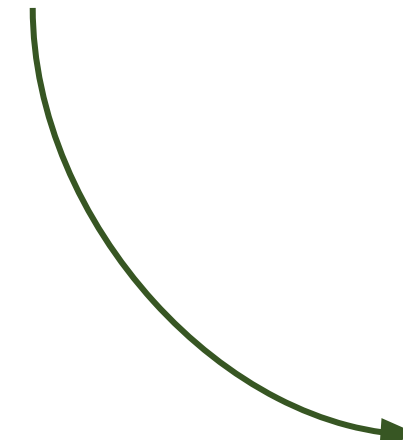
***Practical
ecodesign***



***Industrial
symbiosis***



***Value chain
formalization***



Conclusions

- The plastics sector in Colombia is dominated by packaging and intermediate transformation activities.
- The main barriers to implementing circular economy practices include high costs of recycled materials, insufficient and informal recycling infrastructure.
- Technology-transfer barriers originate from minimal university–industry interaction.
- To enhance alignment between academia and industry is required structured collaboration platforms, targeted capacity-building programs, and projects focused on applied research and ecodesign.



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Thank you for your attention!

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